

National Academy of Science and Technology

(Affiliated to Pokhara University)

Dhangadhi, Kailali

Pre-University Examination

Level: Bachelor
Program: B.E. Computer
Course: Software Engineering

Semester : V_Fall

Year : 2025

F.M. : 100

P. M. : 45

Time : 3 Hrs.

Candidates are required to give their answer in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Suppose a Suppose that a Basic project was estimated to be 400 KLOC (kilo lines of code). Calculate effort and time for each of the three modes of development. 7
b) Consider a software project with the following information 8
domain characteristic for the calculation of function point metric.
Number of external inputs (I) = 25
Number of external output (O) = 55
Number of external inquiries (E) = 33
Number of files (F) = 09
Number of external interfaces (N) = 04
It is given that the complexity weighting factors for I, O, E, F, and N are 4, 5, 4, 10, and 7, respectively. It is also given that, out of fourteen value adjustment factors that influence the development effort, four factors are not applicable, each of the other four factors has value 3, and each of the remaining factors has value 4. The computed value of the function point metric is ____.
2. a) Analyze the Scrum framework in detail. How does Agile Project Management differ from traditional approaches when scaling to large, multi-team environments? 8
b) Describe the Software Development Lifecycle (SDLC) phases. How does the Prototyping model assist in requirements validation compared to the Incremental model? 7

3. a) Using the principles of Object-Oriented Analysis, model a library management system Using Use Case diagram and explain how Domain Analysis is important. 7
- (b) Compare Behavioral Models and Data Models. Create a Context Model (Level 0 and Level 1 DFD) for an E-commerce system. 8
4. a) Discuss the Design Process and its guiding principles. How does "Security by Design" change the architectural approach to software development? 8
- b) Explain the importance of Design Patterns in modern software engineering. How do they contribute to software reuse and maintainability? 7
5. a) Differentiate between Validation and Verification. Explain the roles of Alpha, Beta, and Acceptance testing in the user testing phase. 8
- b) Discuss the Software Evolution process. Why is Reengineering often more cost-effective than developing a new system from scratch? 7
6. a) What is Software Quality Assurance (SQA)? Explain the difference between Reviews and Inspections in the context of quality management. 8
- b) Explain the concept of Software Reuse through Application Frameworks and COTS (Commercial Off-The-Shelf) products. What are the primary risks of COTS reuse? 7
7. Write short notes on following (Any Two) 5x2
- a) Functional Vs Non-functional requirements.
 - b) Software Quality Assurance (SQA)
 - c) Waterfall Vs RAD model

Good Luck!!!